

Table C.22: Model Selection: Abnormal

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
AUTOMATION BIAS (b)	0.11 (0.03)	0.06 (0.02)	0.10 (0.03)	0.11 (0.03)	0.06 (0.02)	0.06 (0.02)	0.08 (0.02)	0.06 (0.02)	0.08 (0.03)	0.08 (0.03)	0.06 (0.02)	0.06 (0.02)
OWN INFO BIAS (d)	1.06 (0.02)	1.07 (0.02)	1.06 (0.02)	1.07 (0.02)	1.07 (0.02)	1.07 (0.02)	1.06 (0.02)	1.07 (0.02)	1.06 (0.02)	1.07 (0.02)	1.07 (0.02)	1.07 (0.02)
CONSTANT	0.38 (0.07)	0.29 (0.06)	0.36 (0.07)	0.39 (0.07)	0.30 (0.06)	0.29 (0.06)	0.34 (0.06)	0.29 (0.06)	0.34 (0.07)	0.34 (0.07)	0.29 (0.06)	0.29 (0.06)
FOCAL s_A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
OTHER s_A		✓			✓	✓		✓			✓	✓
FOCAL s_E			✓	✓	✓	✓			✓	✓	✓	✓
OTHER s_E				✓		✓				✓		✓
CLINICAL HISTORY s_E							✓	✓	✓	✓	✓	✓
J-STATISTIC	12.65	15.83	13.55	12.74	15.0	15.43	15.8	15.95	16.19	14.87	15.78	15.41
MMSC-BIC	-25.33	-22.14	-24.43	-25.24	-22.98	-22.54	-22.17	-22.03	-21.78	-23.1	-22.2	-22.56
NUMBER OF MOMENTS	12	12	12	12	12	12	12	12	12	12	12	12
Observations	5710	5710	5710	5710	5710	5710	5710	5710	5710	5710	5710	5710
R-SQUARED	0.37	0.32	0.34	0.35	0.32	0.32	0.34	0.32	0.33	0.33	0.32	0.32

Note: This table presents results of the model selection exercise as described in Table 3, though this table only includes Abnormal.

C.5.4 Model Selection Without Adjusting for Measurement Error

This section presents the results of the model selection exercise not accounting for measurement error in the human signal. In these analyses, the instruments are constructed using the radiologist’s report on the case in the treatment arm without AI assistance. Note that some time elapses between the reads, so the radiologist likely observes a different draw of s_E introducing measurement error into the right-hand side variables of equation 7. This is why the preferred method uses instruments constructed using a leave-one-out average of reports for the case. Table C.23 presents results for top-level pathologies with AI, Table C.24 presents results for all pathologies with AI, and Table C.25 presents results for abnormal.

Table C.23: Model Selection: Top Level with AI without Accounting for Measurement Error

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
AUTOMATION BIAS (b)	0.49 (0.02)	0.37 (0.02)	0.56 (0.03)	0.53 (0.03)	0.37 (0.02)	0.38 (0.02)	0.42 (0.02)	0.37 (0.02)	0.48 (0.03)	0.52 (0.02)	0.36 (0.02)	0.38 (0.02)
OWN INFO BIAS (d)	1.00 (0.02)	0.89 (0.02)	0.95 (0.02)	0.95 (0.02)	0.90 (0.02)	0.90 (0.02)	0.93 (0.02)	0.90 (0.02)	0.91 (0.02)	0.94 (0.02)	0.90 (0.02)	0.89 (0.02)
CONSTANT	0.32 (0.04)	0.13 (0.04)	0.29 (0.04)	0.31 (0.04)	0.12 (0.04)	0.13 (0.04)	0.21 (0.04)	0.14 (0.04)	0.20 (0.04)	0.27 (0.04)	0.14 (0.04)	0.12 (0.04)
FOCAL s_A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
OTHER s_A		✓			✓	✓		✓			✓	✓
FOCAL s_E			✓	✓	✓	✓			✓	✓	✓	✓
OTHER s_E				✓		✓				✓		✓
CLINICAL HISTORY s_E							✓	✓	✓	✓	✓	✓
J-STATISTIC	3.47	1.06	2.04	2.94	4.6	1.98	0.01	1.75	0.0	3.7	9.19	0.0
MMSC-BIC	-4.97	-7.38	-2.18	-1.28	-3.84	-6.46	-4.21	-10.91	0.0	-0.52	-7.69	0.0
NUMBER OF MOMENTS	5	5	4	4	5	5	4	6	3	4	7	3
Observations	11420	11420	11420	11420	11420	11420	11420	11420	11420	11420	11420	11420
R-SQUARED	0.58	0.55	0.56	0.56	0.55	0.55	0.56	0.55	0.56	0.56	0.55	0.55

Note: This table presents results of the model selection exercise as described in Table 3 for top-level pathologies with AI assistance without accounting for measurement error in the radiologist reports.

Table C.24: Model Selection: Pooled with AI without Accounting for Measurement Error

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
AUTOMATION BIAS (b)	0.45 (0.02)	0.35 (0.01)	0.45 (0.02)	0.46 (0.02)	0.36 (0.01)	0.35 (0.01)	0.39 (0.02)	0.35 (0.01)	0.41 (0.02)	0.45 (0.02)	0.36 (0.01)	0.35 (0.01)
OWN INFO BIAS (d)	1.01 (0.01)	0.93 (0.01)	0.97 (0.01)	0.98 (0.01)	0.93 (0.01)	0.94 (0.01)	0.96 (0.01)	0.93 (0.01)	0.95 (0.01)	0.97 (0.01)	0.94 (0.01)	0.93 (0.01)
CONSTANT	0.21 (0.04)	0.00 (0.04)	0.10 (0.04)	0.14 (0.04)	-0.01 (0.04)	0.01 (0.04)	0.07 (0.04)	0.00 (0.04)	0.04 (0.04)	0.12 (0.04)	0.01 (0.04)	-0.01 (0.04)
FOCAL s_A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
OTHER s_A		✓			✓	✓		✓			✓	✓
FOCAL s_E			✓	✓	✓	✓			✓	✓	✓	✓
OTHER s_E				✓		✓				✓		✓
CLINICAL HISTORY s_E							✓	✓	✓	✓	✓	✓
J-STATISTIC	11.7	0.0	0.0	1.18	4.21	7.31	0.0	8.13	0.0	0.0	2.85	8.57
MMSC-BIC	-13.62	0.0	0.0	-3.04	-12.67	-9.57	0.0	-8.75	0.0	0.0	-18.25	-8.31
NUMBER OF MOMENTS	9	3	3	4	7	7	3	7	3	3	8	7
Observations	57100	57100	57100	57100	57100	57100	57100	57100	57100	57100	57100	57100
R-SQUARED	0.58	0.56	0.56	0.56	0.56	0.56	0.57	0.56	0.56	0.56	0.56	0.56

Note: This table presents results of the model selection exercise as described in Table 3 for all pathologies with AI assistance without accounting for measurement error in the radiologist reports.